MAE 541 Final Project Proposal: Analysis of Periodic Orbits in the Circular Restricted Three Body Problem

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I. Introduction

Growing international and commercial interest in the cislunar (Earth–Moon) domain is driven by the goal of establishing a sustained human and robotic presence on and near the Moon. As a result, the number of planned missions in this region is increasing rapidly, with some projections estimating that nearly 50 missions to the cislunar environment will be launched by 2030, including efforts to support human exploration infrastructure, resource utilization, and commercial activities [1].

The cislunar environment is governed by complex dynamics, as objects move under the gravitational influence of both the Earth and the Moon. In addition, they are affected by further perturbations, such as the gravitational pull of other bodies, including the Sun and Jupiter, as well as solar radiation pressure. The growing number of missions in this region requires a thorough understanding of its dynamical structure, which can be developed through mathematical analysis of the underlying system.

A common astrodynamics model used in preliminary mission design is the Circular Restricted Three-Body Problem (CR3BP). The CR3BP is a Hamiltonian system that describes the motion of a massless body under the gravitational influence of two primary bodies in a rotating frame of reference. Although it is a simplified model, it still captures many of the essential dynamical features of three-body systems.

Periodic orbits are one particularly important dynamical feature for space missions, which can be studied in the CR3BP. In the Earth-Moon system, such orbits often serve as mission targets, and properties such as their stability and spatial location play an important role in mission design. Low-thrust trajectory design frequently exploits the stable and unstable invariant manifolds of these orbits to approach or depart from them in a fuel-efficient manner. The study of periodic orbits also supports the development of efficient long-term propagation methods for cislunar objects, for example through normal form theory.

II. Proposal

Our project, aims to study the behaviour of the Earth-Moon CR3BP with a particular focus on periodic orbits. The CR3BP describes the motion of a body of negligible mass under the gravitational force of two larger bodies. The larger bodies are referred to as the primary with mass m_1 and the secondary with mass m_2, where $m_1 > m_2$. A system of natural units is used for normalizing position, time and velocity variables: the distance unit (DU) corresponds to the distance between primary and secondary, the time unit (TU) is the orbital period of primary and secondary divided by 2π, and the mass unit (MU) is $m_1 + m_2$. The only parameter of the system is the *mass ratio* $\mu = m_2/(m_1 + m_2)$, which in the case of the Earth-Moon system takes the value: $\mu = 0.012$. The dynamics are described in a rotating frame of reference shown in Fig. II.1, with primary and secondary on the r_1 axis at $r_1 = \mu$ and $r_1 = 1 - \mu$. They are given by

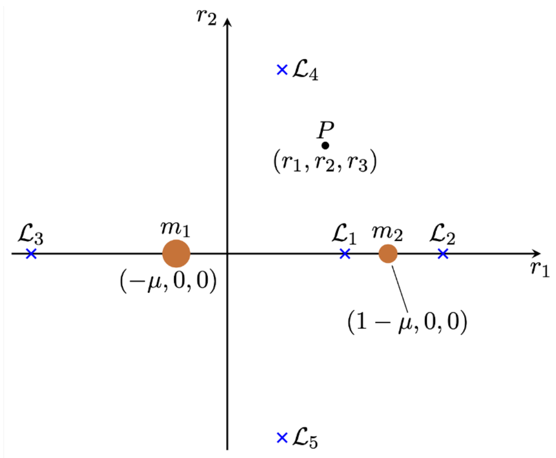

Fig. II.1. The rotating coordinate frame of the CR3BP with the two primaries on the r_1 axis and the 5 libration points.

the vector field

$$\ddot{\boldsymbol{r}} = \begin{pmatrix} 2v_2 + r_1 - (1-\mu)\frac{r_1+\mu}{\rho_1^3} - \mu\frac{r_1-1+\mu}{\rho_2^3} \\ -2v_1 + r_2 - (1-\mu)\frac{r_2}{\rho_1^3} - \mu\frac{r_2}{\rho_2^3} \\ -(1-\mu)\frac{r_3}{\rho_1^3} - \mu\frac{r_3}{\rho_2^3} \end{pmatrix}, \tag{II.1}$$

where $\rho_1 = \sqrt{(r_1+\mu)^2 + r_2^2 + r_3^2}$ and $\rho_2 = \sqrt{(r_1 - 1 + \mu)^2 + r_2^2 + r_3^2}$ are the distances to primary and secondary respectively.

We aim to study this nonlinear system using several tools from dynamical systems theory. In particular, our project will consist of the following components, with the main responsibility for each part indicated in brackets:

- **Equilibrium Points**: The CR3BP has five equilibrium points, commonly referred to as Lagrange or libration points. We will derive their locations, local linearizations of the vector field (II.1), and study their stability as a function of the mass ratio. (*Claudio*)
- **Computing Periodic Orbits**: The CR3BP admits infinitely many periodic orbits [2]. We will focus on planar periodic orbits, in particular resonant orbits and orbits around the libration points. Computing such orbits is not always straightforward because of their abundance and stability properties. We will generate a dataset of periodic orbits using differential correction together with informed initial guesses based on known orbit characteristics. (*Jannik*)
- **Poincaré Map at $y = 0$**: This part of the project focuses on the planar CR3BP (PCR3BP), which has a four-dimensional phase space. Introducing a Poincaré map at $y = 0$ turns periodic orbits of the continuous system into fixed points of the map. This provides a useful tool for visualizing the families of periodic orbits computed in the previous section in a three-dimensional space and for studying how densely they populate different regions of phase space. (*Jack*)
- **Poincaré Map at $y = 0$ for Fixed Energy**: This component also focuses on the PCR3BP and aims to illustrate how the system dynamics depend on the energy level. We will study puncture points on a two-dimensional Poincaré map defined by fixing $y = 0$ at a given energy level. By propagating a large number of initial conditions over long time spans, we aim to visualize how chaotic behavior becomes more pronounced as the energy increases. (*Jack*)
- **Bifurcations of Periodic Orbits**: Families of periodic orbits undergo bifurcations as the mass parameter or orbit energy varies. In this part of the project, we will identify and visualize selected bifurcations within these families. (*Claudio*)
- **Normal Form around Periodic Orbits**: By introducing a sequence of near-identity transformations to normal form coordinates, the dynamics near a periodic orbit can be simplified. This enables more efficient propagation in normal form coordinates, with potential applications to long-term uncertainty propagation. We will apply this approach to an example Lyapunov orbit around one of the libration points. (*Jannik*)

References

[1] K. Johnson, "Fly Me to the Moon: Worldwide Cislunar and Lunar Missions," Center for Strategic and International Studies (CSIS) Aerospace Security Project, Report, Feb. 2022.

[2] H. Poincaré, *Les Methodes Nouvelles de la Mecanique Celeste. 2, Methodes de mm. Newcomb, Gylden, lindstedt et bohlin.* Dover publications, 1957.